

Annotation Guidelines: Multi-Class Vehicle Detection Dataset

1. Project Overview

This project aims to create a high-quality annotated dataset for vehicle detection using dashcam footage from Indonesian roads. The primary goal is to train robust object detection models capable of identifying, localizing, and classifying vehicles into specific sub-types.

Tool Used: Labelbox

Annotation Type: Bounding Box with Nested Attributes

Main Classes: Car, Truck, Bus

Total Sub-Types: 9 distinct categories

2. General Labeling Rules

To ensure data consistency and model accuracy, all annotators must adhere to the following general principles:

- **Tight Bounding Boxes:** The bounding box must tightly enclose the visible parts of the vehicle. Include mirrors, bumpers, and spoilers if visible. Exclude excessive background (road, sky, trees).
 - **Visibility Threshold:** Only annotate vehicles where **>50%** of the body is visible. Vehicles that are heavily occluded or cut off at the image edge should be labeled only if the majority of the structure is identifiable.
 - **Motion Blur:** Annotate vehicles even if they exhibit motion blur, provided the vehicle type can be reasonably identified. If the blur makes identification impossible, skip the object.
 - **No Overlapping Labels:** Do not draw multiple boxes for the same single vehicle. Each distinct vehicle gets exactly one bounding box.
 - **Ignore Small Objects:** Vehicles that appear extremely small (e.g., less than 10x10 pixels) due to distance may be skipped unless they are critical for context.
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3. Class Definitions & Nested Attributes

We utilize a **hierarchical ontology**. First, select the Main Class (Car/Truck/Bus), then select the specific Sub-Type from the nested attribute menu.

Main Class 1: Car (Color: Blue)

Definition: Standard passenger vehicles designed primarily for transporting people (typically < 8 passengers).

Nested Attribute: Car_Type

Select one of the following:

- **Sedan:** Traditional 3-box configuration (engine, passenger, cargo compartments separate).
- **Hatchback:** Compact car with a rear door that swings upward to provide access to a cargo area.
- **SUV (Sport Utility Vehicle):** Higher ground clearance, larger body size, often with 4-wheel drive capabilities.
- **Sport:** Low-profile, aerodynamic design, typically 2-door coupes or high-performance vehicles.
- **Others:** Minivans, MPVs, small personal pickups, or ambiguous passenger vehicles.

■ Main Class 2: Truck (Color: Yellow)

Definition: Large motor vehicles designed for transporting heavy goods, materials, or large cargo.

Nested Attribute: Truck_Type

Select one of the following:

- **Container Truck:** Trucks carrying standardized shipping containers (usually blue, red, or white rectangular boxes).
- **Box Truck:** Trucks with an enclosed cuboid-shaped cargo area attached to the chassis.
- **Dump Truck:** Trucks with an open-box bed used for transporting loose material like sand, gravel, or demolition waste.
- **Pickup Truck:** Light-duty trucks with an enclosed cab and an open cargo area with low sides and tailgate. *(Note: Only label as Truck if it's a commercial/large pickup. Small personal pickups go to Car -> Others).*
- **Others:** Tanker trucks, fire trucks, or specialized utility trucks.

■ Main Class 3: Bus (Color: Green)

Definition: Large motor vehicles designed to carry many passengers (typically > 8 passengers).

Nested Attribute: Bus_Type

Select one of the following:

- **Public City Bus:** Standard buses used for urban public transport, often with multiple doors and standing room.
- **Minibus:** Smaller buses, typically used for short-distance public transport or shuttle services (e.g., Angkot, Elf).
- **School Bus:** Distinctive yellow buses or buses marked for school transport.
- **Others:** Tourist buses, double-decker buses, or ambiguous large passenger vehicles.

4. Handling Edge Cases

Scenario	Action	Reason
Occlusion	Label the visible part tightly. Select sub-type based on visible features.	The model needs to learn partial features.
Reflections	Do NOT label reflections in windows/mirrors.	Reflections are not physical objects.
Posters/Billboards	Do NOT label vehicles in advertisements.	Only label real-world physical vehicles.
Ambiguous Sub-Type	Use best judgment. If unsure between Sedan/Hatchback, choose Others.	Consistency is more important than perfect taxonomy.
Truck vs Pickup	If it has a separate cab and open bed, use Truck -> Pickup Truck. If integrated body, use Car -> Others.	Clear distinction helps model learning.

5. Quality Assurance Checklist

Before submitting each batch of annotations, please verify:

1. Are all bounding boxes tight around the vehicle?
2. Is the correct Main Class (Car/Truck/Bus) selected?
3. Is the correct **Nested Sub-Type** attribute filled for every object?
4. Are there no duplicate boxes for the same vehicle?
5. Are reflections and billboard images ignored?